Francesco Palandra

AI Researcher

palandrafrancesco@gmail.com | frklin.github.io | github.com/Frklin | linkedin.com/in/francescopalandra

A Master's student at the University of Rome La Sapienza, undertaking the Computer Science course. I have always been fascinated by the world of computer science and its potential to transform the way we live and work. In particular, I am intrigued by the research process and the pursuit of solutions to problems that have yet to be solved. I believe that having a clear vision for the future is crucial in any field, especially in the ever-evolving landscape of technology. As a result, I am constantly seeking to broaden my knowledge and understanding of the latest developments in the field, and I am always looking for ways to contribute to the advancement of technology. My research interests lie primarily in the fields of Artificial Intelligence, Computer Graphics and Geometry Computing. Throughout my Master's degree, I have presented numerous innovative and competitive projects that reflect my interest and my potential.

Education

University of Rome La Sapienza — *Rome, Italy*

MSc in Computer Science — *Oct 2022 - present*

- Working as Tutor for Bachelor students
- **Courses:** Data Science, Graphics and Vision, Knowledge Representation and Reasoning, Statistical Theory and Methods, Learning Skills through Case Studies, Artificial Intelligence, Machine Learning, Statistical Learning and Networking
- **GPA:** 29.85/30 (top 1%)

University of Rome La Sapienza — *Rome, Italy*

BCs in Computer Science — *Sept 2019 - Oct 2022*

- **Courses:** Fundamentals of Computer Science and Calculus, Computer Architecture, Algorithms, Data structures, Graphics, Web and Software Architecture, Data Management and Analysis
- **Thesis Title:** Vehicle Type Detection through BLE Device Scans
- **Final Grade:** 108/110

Work Experience

Junior Researcher — *Jan 2024 - Present*

Sapienza University of Rome — *Rome, Italy*

- I am currently working on three different research projects inside my university:
  - 'GS-Edit,' a standout project with the collaboration of Professor Emanuele Rodolà and Geometry, Learning and Applied AI (GLADIA) Lab, leverages Gaussian Splatting for 3D object editing, blending image and text conditioning in a novel way. This work was submitted to ECCV 2024.
  - As a winner of the honorable program, under the supervision of Professor Fabio Galasso, I joined the Perception and Intelligence Laboratory (PINLab) in a Virtual Human project aiming to generate natural human pose sequences inside a scene from a prompt.
  - my junior research fellowship has allowed me to contribute to a NATO-financed drone testbed project, focusing on a novel routing algorithm based on reinforcement learning.
- **Technical Skills:** Pytorch, ML, Documentation, LaTeX.
- **Soft Skills:** Problem Solving, Logical Thinking, Time Management, Teamwork, Adaptability.

Tutor — *Mar 2023 - present*

Sapienza University of Rome — *Rome, Italy*

- Firstly assigned to teach Calculus Unit II, a first-year course in the Computer Science program that covers fundamental concepts of calculus, with a focus on integrals and differential equations. Responsible for instructing both English and Italian version of the course, with a classroom size of 40 students per class. Conducted exercises, addressed questions, and prepared students for their final exams.
- Additionally, I was entrusted with the responsibility of instructing courses on algorithms and programming to undergraduate students in the Computer Science program. My objective in this role was to facilitate comprehensive understanding of both theoretical and practical aspects of programming. This included strengthening students' grasp of algorithmic principles, fostering their problem-solving skills, and cultivating their capacity to apply these concepts effectively in a practical context.
- **Technical Skills:** Math, Classroom Management, LaTeX.
- **Soft Skills:** Communication, Logical Thinking, Time Management, Teamwork, Adaptability.

Software Developer — *May 2022 - Jan 2024*

Sapienza University of Rome — *Rome, Italy*

- As part of a team of three, I developed a model utilizing Bluetooth Low-Energy technology to predict the specific vehicle in which the user is located. This model was designed with the primary aim of improving the efficiency of the smart parking system, and it will be integrated into a smart parking application made by the Gamification Lab at the Department of Computer Science at the University of Rome La Sapienza.
- An application has been designed and developed in Swift to illustrate the various functional components.
- After my graduation I continued to work in the team leading a group of five Master's and PhD students aiming for a patent request.
- **Technical Skills:** Swift, Go, Git.
- **Soft Skills:** Leadership, Teamwork, Problem Solving, Time Management, Communication, Logical Thinking.

University Projects

MetaFusion - A forgeable NFT Collection
Best Project

Blockchain and Distributed Ledger Technologies (MSc) *2024*

- Metafusion is DApp that allows users to generate images harnessing the power of Stable Diffusion. A fully functional website has been developed in SvelteKit that allows users to exchange and forge NFTs.
- A hardhat node simulates the blockchain while an oracle and a tracker create and handle the NFTs creation and exchange.
- **Technical Skills:** Solidity, Svelte, SQLlite.
- **Soft Skills:** Leadership, Teamwork, Documentation, Presentation skills, Report writing.

GS Edit - 3D scene editing through 2D diffusion model

Deep Learning and Applied Artificial Intelligence (MSc) *2023*

- a novel approach that is capable of editing 3D objects based on image and text conditioning. This unlocks the possibility to edit the style and appearance of 3D objects without altering their main details.
- We tackle the problem by leveraging 2D diffusion models to generate variations and Gaussian Splatting to build the 3D model from those variations.
- This work has been submitted to EECV 2024
- **Technical Skills:** Pytorch, Transformers, OpenGL.
- **Soft Skills:** Leadership, Teamwork, Documentation, Presentation skills, Report writing.

Tweepic - Twitter Clustering
Best Project

Big Data Computing (MSc) *2023*

- Tweepic is a sophisticated neural network specifically designed to categorize and analyze tweets on Twitter. This system becomes particularly effective during instances of a 'tweet boom,' a phenomenon characterized by a massive influx of tweets following a specific event. Upon identification of a tweet boom, the relevant tweets are integrated into a processing pipeline that extracts embeddings using a multilingual, pretrained model known as XMLRoberta. Subsequently, a graph is constructed, where the weights of the edges are determined by word, hashtag, and sentence embeddings. A classifier is then employed to discern and sever the edges linking tweets pertaining to disparate topics. Ultimately, this process results in the isolation of connected components, each representing a unique topic of discussion during the tweet boom..
- This project has been assigned to a PhD student as his PhD thesis
- **Technical Skills:** PySpark, Numpy, Transformers, Pytorch, PySparkNLP.
- **Soft Skills:** Leadership, Teamwork, Time Management, Presentation skills, Documentation.

Virtual Closet - Smart Mirror
Best Project

Human Computer Interaction (MSc) *2023*

- In collaboration with a team of four, we engineered a 'smart mirror,' a virtual fitting room where users can virtually try on clothes. The interface was constructed utilizing fundamental HTML/CSS programming, augmented with MediaPipe to track facial and hand landmarks. This integration enabled user interaction with the mirror via hand gestures. Since the objective was to project only clothes and accessories onto the mirror, sans user image, we had to develop a proprietary positional system from the ground up. This system, built with three.js, allowed for the clothing to rotate synchronously with the user.
- **Technical Skills:** Three.js, HTML/CSS, RasberryPi, Figma, Mediapipe, Arduino.
- **Soft Skills:** Leadership, Teamwork, Presentation skills, Creativity, Design.

UAVs patrolling through a Genetic Q-learning Algorithm

Internet of Things (MSc) *2023*

- In our quest to devise a solution for the issue of drone patrolling within specific time windows, we introduced a novel methodology that merges a random and clustering-based population generated by a multi-objective genetic algorithm (NSGA II) with a population established by a Q-learning algorithm. The Q-learning algorithm endeavors to discover the optimal path, thereby contributing to the efficiency and effectiveness of drone patrolling operations.
- **Technical Skills:** ROS, Gazebo, Python, Deap
- **Soft Skills:** Documentation, Creativity, Teamwork, Presentation skills, Report writing.

Person ReID with Few Shot Learning

Biometric System (MSc) *2023*

- I led a group of master students and developed an advanced model for person re-identification (ReID) with Few-Shot Learning utilizing Locally Aware Transformers. The developed model has a competitive edge in terms of its state-of-the-art performance.
- Conducted a comprehensive analysis of the state of the art in Few Shot Learning Tracking, reviewing and comparing existing techniques and approaches, and identifying gaps and opportunities for future research.
- Submitted the paper to the International Joint Conference on Biometrics (IJCB) held in Ljubljana, with the aim of presenting and discussing the proposed method with a broader audience of researchers and practitioners in the field of biometrics and computer vision.
- **Technical Skills:** Numpy, Transformers, Pytorch, Pandas, scikit-learn.
- **Soft Skills:** Leadership, Teamwork, Time Management, Presentation skills, Report writing.

QMR and Q-FANET algorithm

Autonomous Networking (MSc) *2023*

- Developed a Q-learning based Multi-objective optimization Routing protocol (QMR) for Flying Ad Hoc Networks (FANETs) of drones, which aims to minimize packet delivery delay and drone energy consumption. The QMR algorithm is implemented as a three-block system that involves location acquisition, routing neighbor discovery, and routing decision, each with a specific function and role in the overall protocol.
- Implemented the QMR algorithm in a simulator environment, allowing for the evaluation of its performance under various scenarios and conditions. The simulator already included the location acquisition and routing neighbor discovery modules, while the QMR algorithm was developed and integrated into the routing decision block.
- Conducted a comparative evaluation of the QMR algorithn with existing state-of-the-art algorithms for FANET routing in order to demonstrate the efficiency and effectiveness of the algorithm in minimizing packet delivery delay and drone energy consumption.
- **Technical Skills:** Python, Network Algorithms, LaTeX
- **Soft Skills:** Time Management, Teamwork, Presentation skills, Report writing.

Physically-based Renderer

Fundamentals of Computer Graphics (MSc) *2022*

- Implemented a physically-based renderer with ray tracing, which allows for accurate and realistic simulations of light behavior in a virtual scene, including features such as reflections, refractions, and shadows.
- Included support for volumetric media rendering, allowing for the creation of realistic effects such as fog, smoke, and clouds, through the use of advanced techniques such as null scatter path integral formulation
- The renderer was developed using C++ and the Yocto/GL library, which enabled the implementation of both the rendering pipeline and the simulation foundations using mass-spring and position-based dynamics (PBD).
- **Technical Skills:** NanoVDB, OpenGL, YoctoGL, C++.
- **Soft Skills:** Problem Solving, Time Management, Documentation, Creativity.

Minor Projects

2024	Zero-1P2P-3 for 3D Object Editing	*Python*
2024	**(Best Project)** Relation Extraction model for MNLP	*Python*
2024	Word Sense Disambiguation model for MNLP	*Python*
2023	Event Detection for MNLP	*Python*
2022	Machine Learning Model for ASL recognition	*Python*
2022	Social Network Application for Sapienza students	*Swift*
2022	Website for traveling route suggestions	*HTML/CSS*
2021	Chat-room	*C*
2020	API for a NLP website	*Java*
2020	Database for a Covid Hotel	*SQL*

Grants and Fellowship

2024	**First** Honorous Programme Scholarship at Sapienza University of Rome	*Rome, Italy*
2024	**Winner** Oxford Machine Learning Summer School in Generative AI	*Oxford, UK*
2024	**Winner** Junior Research Scholarship for Testbed of Drone Networking	*Rome, Italy*
2023	**Winner** Programming and Algorithms Tutor Fellowship at Sapienza University of Rome	*Rome, Italy*
2023	**Winner** Scholarship for Billion User Cloud Application Summer School by Google	*Como, Italy*
2023	**Winner** Calculus II (both English and Italian) Tutor Fellowship at Sapienza University of Rome	*Rome, Italy*

Skills

Programming	Python (Pandas, PyTorch, NumPy, Scikit-learn. etc.), Python, C, C++, Java, Swift, R, Matlab, Assembly, MySQL, PostgreSQL, Spark.
Web	HTML, CSS, Javascript, PHP.
Miscellaneous	Linux, Shell, LaTeX, Git, OpenCV, Microsoft Office.
Soft Skills	Time Management, Teamwork, Problem-solving, Creativity, Documentation, Engaging Presentation.

Languages

Italian	**C2** - Native Speaker
English	**C1** - TOEFL iBT Certificate
French	**B1** - Middle School Class

Interests

Music	Since I was a kid I have always been into music, I've been playing piano for 13 years and I am currently a lead singer in a rock band.
Sport	I've played various sports including football, swimming, and triathlon, and now I enjoy playing tennis.
Cooking	I love Italian-style cooking, especially Roman cuisine, and enjoy experimenting with new recipes.
Art	Drawing has been a hobby of mine since childhood, and my mother, who is an artist, taught me the basics of charcoal drawing.
Astronomy	I find Physics and Astronomy fascinating and I like to read Stephen Hawking's books on universe and black holes in my free time.
Cinema	I appreciate the art of cinematography and love analyzing movies, particularly those by Christopher Nolan.
Traveling	My friends and I enjoy exploring small and lesser-known towns during weekends, and admiring the beauty of Renaissance Italy.
Photography	With my father we share the hobby of the photography. When I was around the age of 13 he used to bring me at parks and photograph squirrels.